



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

ENTHUSIAST ADVANCE COURSE

PHASE : MEB-I (A)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 04-03-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	4	2	1	1	3	3	4	3	4	3	3	4	1	4	2	1	1	3	3	1	3	4	3	3	4	1	1	1	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	1	4	2	4	4	4	2	2	2	3	3	2	1	1	3	3	2	2	2	3	2	3	4	1	3	4	1	3	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	2	4	4	3	2	2	4	2	3	1	2	3	3	2	4	4	3	4	1	1	4	4	3	1	4	2	2	3	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	4	3	1	3	2	1	1	1	3	3	4	2	2	2	2	4	2	3	4	4	3	4	3	2	3	3	3	2	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	3	4	3	3	3	2	2	3	2	2	1	1	1	3	4	3	4	4	3	1	3	2	2	4	4	1	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	2	1	3	2	3	2	2	4	1	3	1	4	3	1	3	4	4	2	4	2	3	2	2	1	1	3	4	2	4
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	2	4	1	3	1	1	4	4	4	3	4	2	3	1	2	4	1	2	3	3										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

1. **Ans (1)**

New charges will be, $-2\mu\text{C}$ and $+3\mu\text{C}$.

$$\text{In first case ; } 40 = \frac{1}{4\pi\epsilon_0} \times \frac{3 \times 8}{r^2}$$

$$\text{In second case ; } F = \frac{1}{4\pi\epsilon_0} \times \frac{(-2 \times 3)}{r^2}$$

$$\therefore \frac{F}{40} = \frac{-2 \times 3}{3 \times 8}$$

$$\text{or } F = -10 \text{ N}$$

2. **Ans (4)**

$$\Delta\phi = 4 \times 10^3 - 8 \times 10^3$$

$$= -4 \times 10^3 \text{ N-m}^2/\text{coulomb}$$

According to Gauss's law,

$$q = -4 \times 10^3 \epsilon_0 \text{ coulomb}$$

6. **Ans (3)**

The electric flux through any surface is equal to the product of electric field intensity at the surface and component of the surface perpendicular to electric field $= E \times \pi R^2 = \pi R^2 E$.

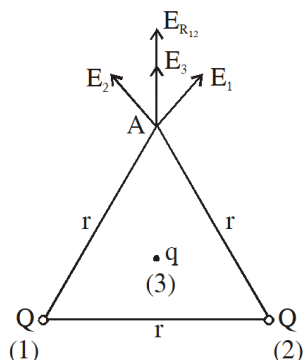
9. **Ans (3)**

$$\phi_{S1} = \frac{\epsilon q}{\epsilon_0} = \frac{Q/2}{\epsilon_0} = \frac{Q}{2\epsilon_0}$$

$$\phi_{S2} = \frac{\epsilon q}{\epsilon_0} = \frac{Q/2 + Q/4}{\epsilon_0} = \frac{3Q/4}{\epsilon_0} = \frac{3Q}{4\epsilon_0}$$

$$\frac{\phi_{S1}}{\phi_{S2}} = \frac{Q/2 \epsilon_0}{3 Q/4 \epsilon_0} = \frac{4}{3 \times 2} = \frac{2}{3} = \frac{\sqrt{4}}{3}$$

10. Ans (4)



$$E_A = E_{R12} + E_3 = 0$$

$$\Rightarrow \sqrt{3}E + E_3 = 0$$

$$\Rightarrow \sqrt{3} \frac{kQ}{r^2} + \frac{kq}{(r/\sqrt{3})^2} = 0$$

$$\sqrt{3} \frac{kQ}{r^2} = -\frac{3kq}{r^2}$$

$$\boxed{q = -Q/\sqrt{3}}$$

11. Ans (3)

$$E = \frac{F}{q} = \frac{[MLT^{-2}]}{[AT]}$$

$$E = [MLT^{-3}A^{-1}]$$

15. Ans (4)

From Gauss's law flux crossing through closed surface is depending only on charge enclosed by it and not on their location or on the shapes and size of gaussian surface.

16. Ans (2)

According to Gauss's law, the flux of the electric field through a closed loop is given by

$$\phi = \oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0}$$

ϕ will be zero if $E = 0$ or $\vec{E} \perp d\vec{s}$

or $\oint \vec{E} \cdot d\vec{s} = 0$ and in all cases $q_{in} = 0$.

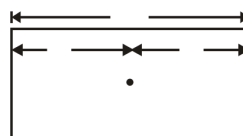
18. Ans (1)

Whatever flux is crossing the conical surface the same amount of flux will enter through its base so

$$\phi = \vec{E} \cdot \vec{A} = E\pi r^2 \quad [\text{where } A = \pi r^2 \hat{n}]$$

19. Ans (3)

$$d_{AP} = \frac{d_{AC}}{\mu}$$



$$\therefore d_1 = 6\mu$$

$$d_2 = 4\mu$$

$$d = d_1 + d_2 = 10\mu = 10 \times 1.5 = 15 \text{ cm}$$

20. Ans (3)

$$f = +0.6 \text{ m}$$

$$v = +0.3 \text{ m}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{0.6} = \frac{1}{0.3} + \frac{1}{u}$$

$$\Rightarrow -\frac{1}{0.6} = \frac{1}{u} \Rightarrow u = -0.6 \text{ m}$$

21. Ans (1)

$$\frac{1}{f} = (\mu - 1) \left(\frac{2}{R} \right) \Rightarrow \frac{3}{2R} = (\mu - 1) \left(\frac{2}{R} \right)$$

$$\mu = 1.75$$

22. Ans (3)

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{P_1}{100} + \frac{P_2}{100} = \frac{1}{100}$$

$$\Rightarrow f = 100 \text{ cm}$$

$\therefore$ A convergent lens of focal length 100 cm.

23. Ans (4)

$$\text{Image by first lens : } \frac{1}{v_1} - \frac{1}{-30} = \frac{1}{10}$$

$$\frac{1}{v_1} = \frac{1}{10} - \frac{1}{30} = \frac{2}{30}$$

$$v_1 = 15 \text{ cm}$$

$$\text{Image by second lens : } u = (15 - 5) = 10 \text{ cm}$$

$$\frac{1}{v_2} - \frac{1}{10} = \frac{1}{-10} \Rightarrow v_2 = \infty$$

$$\text{Image by third lens : } u = \infty$$

$$\frac{1}{v_3} - \frac{1}{\infty} = \frac{1}{30} \Rightarrow v_3 = 30 \text{ cm}$$

24. Ans (3)

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin \frac{A}{2}} \Rightarrow \sqrt{3} = \frac{\sin\left(\frac{60^\circ+\delta_m}{2}\right)}{\sin \frac{60^\circ}{2}}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \sin\left(30^\circ + \frac{\delta_m}{2}\right) \Rightarrow \delta_m = 60^\circ$$

25. Ans (3)

$$d = \frac{t}{\cos r} \sin(i - r)$$

For small angles $\sin(i - r) \approx i - r$; $\cos r \approx 1$

$$d = t(i - r), d = it \left[1 - \frac{r}{i} \right]$$

26. Ans (4)

For water lens

$$\frac{1}{F'} = \left(\frac{4}{3} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$



$$\therefore \frac{1}{F'} = \frac{1}{3} \left(\frac{1}{-R_1} - \frac{1}{R} \right) \dots\dots (i)$$

For glass lens

$$\frac{1}{f} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{R} - \frac{1}{-R} \right) \dots\dots (ii)$$

(i)/(ii)

$$\frac{\frac{1}{f'}}{\frac{1}{f}} = -\frac{\frac{1}{3}}{\frac{1}{2}} \Rightarrow F' = -\frac{3f}{2}$$

 $\therefore$ for 3 lens combination;

$$\frac{1}{f} = \frac{1}{f} + \frac{1}{f} + \frac{1}{-3f/2}$$

$$\Rightarrow F = \frac{3f}{4}$$

27. Ans (1)

$$\frac{(v_B)_{act}}{(v_B)_{app.}} = \frac{\mu_1}{\mu_2} \Rightarrow \frac{-12}{(v_B)_{app.}} = \frac{1}{4/3}$$

$$(v_B)_{app.} = -16 \text{ m/s}$$

$$v_{B/F} = (v_B)_{app} - (v_F)_{act}$$

$$v_{B/F} = -16 \text{ m/s} + 24 \text{ m/s}$$

$$v_{B/F} = 8 \text{ m/s upwards}$$

28. Ans (1)

Due to 9 cm slab; Rise in image

$$y = d - d' = d - \frac{d}{\mu} = d \left[1 - \frac{1}{\mu} \right]$$

$$= 9 \left[1 - \frac{1}{3/2} \right]$$

$$= 9 \left[1 - \frac{2}{3} \right] = 9 \left[\frac{1}{3} \right] = 3 \text{ cm}$$

29. Ans (1)

$$\sin \theta_c = \frac{\sqrt{3}}{2} \Rightarrow \theta_c = 60^\circ$$

30. Ans (1)

$$\frac{w_1}{w_2} = \frac{2}{3} = \frac{-f_1}{f_2}$$

$$f_1 = -2/3 f_2$$

$$\frac{1}{f_{net}} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{120} = \frac{-3}{2f_2} + \frac{1}{f_2}$$

$$\frac{1}{120} = \frac{-3 + 2}{2f_2} = \frac{-1}{2f_2}$$

$$f_2 = -60 \text{ cm}$$

$$f_1 = 40 \text{ cm}$$

32. Ans (1)

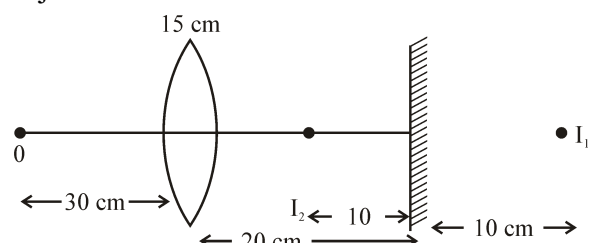
$$\frac{f_1}{f_a} = \frac{{}_a\mu_g - 1}{{}_1\mu_g - 1} = \frac{1.5 - 1}{\frac{1.5}{1.75} - 1} = \frac{1.75 \times 0.50}{0.25} = -3.5$$

$$\therefore f_1 = -3.5 f_a \Rightarrow f_1 = +3.5 R (\because f_a = R)$$

Hence on immersing the lens in the liquid, it behaves as a converging lens of focal length 3.5R.

33. Ans (4)

Initial object is at 2ff from convex lens so first image would also be formed at 2f. i.e. 30 cm from lens. (I_1) it will behave as virtual object to plane mirror at 10 cm distance so plane mirror will form real image (I_2) at 10 cm in front of it. this image will behave as real object to lens at 10 cm.



For Final image from lens

$$u = -10 \text{ v} = ? \text{ t} = 15 \text{ cm}$$

$$v = \frac{uf}{u + f} = \frac{-10 \times 15}{-10 + 15} = -30 \text{ cm}$$

Final image will be virtual at 30 cm from lens that means 10 cm from plane mirror.

34. **Ans (2)**

$$P_{\text{net}} = P_1 + P_2 - dP_1P_2$$

$$= 4 + 6 - \frac{1}{6} \times 4 \times 6$$

$$P_{\text{net}} = +6D$$

35. **Ans (4)**

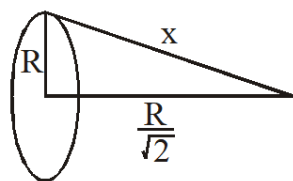
$$h_0 = \sqrt{h_1 h_2}$$

$$3 = \sqrt{9 \times h_2}$$

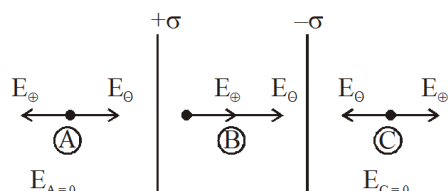
$$h_2 = 1 \text{ cm}$$

SECTION-B37. **Ans (4)**

To cover the charge we need 12 surface

$$\phi = \frac{q}{12\epsilon_0}$$
38. **Ans (2)**

$$x = \sqrt{R^2 + \frac{R^2}{2}} = \sqrt{\frac{3}{2}} R$$

39. **Ans (2)**

$$E_A = 0$$

$$E_B = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0}$$

$$E_B = \frac{2\sigma}{2\epsilon_0}$$

$$E_B = \frac{\sigma}{\epsilon_0}$$

40. **Ans (2)**

$$Q_{\text{net}} = Q_{\text{sys}} = 2Q + (-4Q) + 10Q = 8Q$$

43. **Ans (2)**

$$\text{From graph, slope} = \tan\left(\frac{2\pi}{10}\right) = \frac{\sin r}{\sin i}$$

$$\text{Also } \Rightarrow \mu_2 = \frac{\mu_1}{\sin i} = \frac{1}{\tan\left(\frac{2\pi}{10}\right)} = \frac{4}{3} \Rightarrow \mu_2 > \mu_1$$

It means that medium 2 is denser medium. So total internal reflection cannot occur.

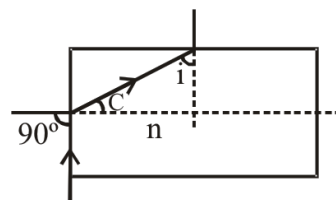
44. **Ans (1)**

$$\mu = \frac{\text{Real depth}}{\text{apparent depth}} = \frac{120}{80} = 1.5$$

45. **Ans (1)**

$$\sin C = \frac{1}{n}$$

For T.I.R. on lateral surface
 $i > C$



$$(90^\circ - C) > C$$

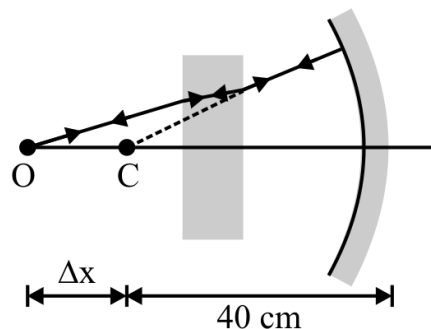
$$\Rightarrow 45^\circ > C$$

$$\sin 45^\circ > \sin C$$

$$\frac{1}{\sqrt{2}} > \frac{1}{n} \Rightarrow n > \sqrt{2}$$

46. **Ans (3)**

$$\Delta x = \left(1 - \frac{1}{\mu}\right) t = \left(1 - \frac{1}{1.5}\right) \times 6$$



$$= 2 \text{ cm.}$$

Distance of object from mirror = 42 cm.

47. **Ans (3)**

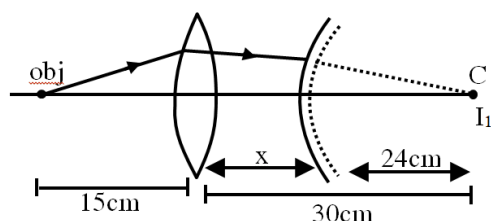
Next power becomes less negative $\therefore$ focal length increases

48. **Ans (2)**

Intensity in AB will smaller & in AC & BD, intensity will be greater.

49. **Ans (2)**

For $D < 4f$ perfect image does not form $40 < 4f$
 $f > 10 \text{ cm}$

50. **Ans (2)**

$$x = 30 - 24 = 6 \text{ cm}$$

SUBJECT : CHEMISTRY**SECTION-A**68. **Ans (4)**

$$\chi_B = \frac{p_A^0 - p_{\text{total}}}{p_{\text{total}}} = \frac{n_B}{n_A}$$

$$\frac{3/(M_w)_B}{19.5/78} = \frac{800 - 760}{760} = \frac{40}{760}$$

$$(M_w)_B = \frac{3 \times 760 \times 78}{19.5 \times 40} = 228 \text{ g}$$

69. **Ans (2)**

$$\pi = iCRT$$

$$15 = i \times 0.4 \times \frac{1}{12} \times 300 = i \times 0.4 \times 25 = 10i$$

$$i = \frac{15}{10} = 1.5 = 1 + 2$$

$$\alpha = 0.5$$

70. **Ans (3)**

$$\pi_1 = \pi_2$$

$$C_1 = C_2$$

$$\frac{10}{60} = \frac{6}{M} \Rightarrow M = 36 \text{ g}$$

71. **Ans (1)**

$$X_{\text{solute}} = 0.2$$

$$x_{\text{solvent}} = 0.8$$

$$\frac{n_{\text{solute}}}{n_{\text{solvent}}} = \frac{1}{4}$$

72. **Ans (2)**

$$\text{Henry's law } P = K_H X$$

$$2 = 5X$$

$$0.4 = X$$

73. **Ans (3)**

$$x_A = 1 \text{ [pure A]}$$

$$P_A^0 = 180 + 90$$

$$= 270 \text{ mm}$$

$$x_B = 1, x_A = 0 \text{ [pure B]}$$

$$P_B^0 = 180 \times 0 + 90$$

$$= 90 \text{ mm}$$

$$P_A^0 : P_B^0 = 270 : 90 = 3 : 1$$

74. **Ans (3)**

Solute + solvent $\rightleftharpoons$ solution

Forward reactions is dissolution

Back ward reaction is crystallisation

75. **Ans (2)**

$$\Delta T_b \propto i \times M$$

76. **Ans (4)**

CH_3OH and $\text{C}_2\text{H}_5\text{OH}$ is not an example of maximum boiling azeotrope.

81. **Ans (1)**

$$\pi = iCRT = 3 \times \frac{0.1 \times 300}{2.5 \times 12} = 3 \text{ atm}$$

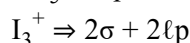
85. **Ans (1)**

$$\pi = iCRT = 1 \times \frac{0.2 \times 1000}{500} \times \frac{1}{12} \times 300$$

$$= 10 \text{ atm}$$

SECTION-B86. **Ans (4)**91. **Ans (4)**

PH_5 not possible IBr_7 not exist due to crowding



= bent

94. **Ans (1)**

$$\pi = icRT$$

$$\pi \propto (i \times c)$$

95. **Ans (3)**

$$K_b = \frac{\Delta T_b}{m} = \frac{0.1 \times 100}{\frac{1.8}{180} \times 1000} = 1K/m$$

96. **Ans (2)**

10% by mass solution means

= 10 g NaOH in 100 g solution

$$n_{NaOH} = \frac{10}{40}; \quad V_{solution} = \frac{100}{2} ml$$

$$M = \frac{n_{NaOH}}{V_{Solution}(in ml)} \times 1000 = \frac{0.25}{50} \times 1000$$

$$= 5$$

97. **Ans (1)**

$$P_A^\circ = P, P_B^\circ = 3P$$

$$X_A = \frac{1}{4}, \quad X_B = \frac{3}{4}$$

$$P_{Total} = P_A^\circ X_A + P_B^\circ X_B = P \times \frac{1}{4} + 3P \times \frac{3}{4}$$

$$= 2.5 P$$

$$P_A^\circ X_A = P_{Total} Y_A$$

$$Y_A = \frac{P \times \frac{1}{4}}{2.5P} = 0.1$$

98. **Ans (1)**

$$M_{final} = \frac{500 \times 2 + \frac{12 \times 10}{60} \times 500}{1000} = 2.5$$

99. **Ans (1)**

$$\text{Solubility} \propto \frac{1}{K_H}$$

100. **Ans (3)**

$$X_A P_A^\circ = Y_A P_S$$

$$\frac{P_A^\circ}{P_S} = \frac{Y_A}{X_A} > 1$$

SUBJECT : BIOLOGY-I**SECTION-A**116. **Ans (3)**

NCERT-Pg#152 (E), 163(H)

117. **Ans (3)**

NCERT-XII Pg. # 149, 150

118. **Ans (3)**

NCERT Pg # 147-149

119. **Ans (2)**

NCERT-XII, Pg. # 158, 159

120. **Ans (3)**

NCERT-XII, Pg. # 147, 148, 149, 150

121. **Ans (2)**

NCERT Pg. # (E) 149 (H) 160

122. **Ans (2)**

NCERT Pg. # 158, 159

123. **Ans (3)**

NCERT-XI, Pg#150,151(E), 162,163(H)

124. **Ans (4)**

NCERT-Pg#153 (E), 165(H)

125. **Ans (3)**

NCERT Pg # 148

SECTION-B142. **Ans (1)**

NCERT Pg. # 154

143. **Ans (3)**

NCERT- 147,149(E), 158,160(H)

SUBJECT : BIOLOGY-II**SECTION-A**163. **Ans (4)**

NCERT Pg # 140

165. **Ans (1)****NCERT Reference:**

Zoology-XII, Page No. 135

166. **Ans (3)**

Pteranodon

167. **Ans (4)**

NCERT-XII Pg# 133, Figure-7.6

168. **Ans (4)**

NCERT-XII, Pg. # 139, Fig.-7.10

169. **Ans (2)**

NCERT (XII) Pg. # 140

170. **Ans (4)**

NCERT(XII) Pg# 132 Para: 7.3

171. Ans (2)

NCERT XII, Page # 138,139 ; Para last & first

172. Ans (3)

NCERT (XII) Pg. # 131

173. Ans (2)NCERT XIIth Pg.#134**174. Ans (2)**

NCERT XII Pg. # 130 & 131

175. Ans (1)

NCERT Pg. # 129

176. Ans (1)

NCERT XII, Pg. No. 134

177. Ans (3)

NCERT (XII) Pg # 131 Para 7.3

178. Ans (4)NCERT (XIIth) Pg. # 64**179. Ans (2)**

NCERT-XII Page No. # 38, 39

180. Ans (4)

NCERT-XII Page No. # 37, 38, 39

181. Ans (2)

NCERT-XII Page No. # 28, 29, 31

182. Ans (4)

NCERT-XII Page No. # 26

183. Ans (1)

NCERT-XII Page No. # 35

184. Ans (3)

NCERT-XII Page No. # 5 – 8

185. Ans (1)

In Potato Axillary bud, in Agave, Bulbils help in Vegetative propagation.

NCERT PAGE: 7

SECTION-B**187. Ans (4)**

NCERT-XII # Page No. 54

191. Ans (4)

NCERT-XII, Pg.#130,131

193. Ans (3)NCERT XIIth, Pg. # 127**194. Ans (1)**

NCERT Pg # 135

196. Ans (4)

NCERT Pg.# 137

197. Ans (1)

Genetic drift operates in small isolated inbreeding population.

199. Ans (3)

NCERT-XII Page No. # 24

200. Ans (3)

NCERT-XII Page No. # 5